

- ③ a Copy and complete Table 6.9 to give values of $1 + 2\sin\theta$ for $0^\circ \leq \theta \leq 360^\circ$ in intervals of 30° .

θ	0°	30°	60°	90°	120°	...	360°
$\sin\theta$	0	0.5	0.87	1	0.87	...	0
$2\sin\theta$	0	1	1.74	2	1.74	...	0
$1 + 2\sin\theta$	1	2	2.74	3	2.74	...	1

Table 6.9

- b Using scales of 1 cm to 30° on the horizontal axis and 2 cm to 1 unit on the vertical axis, draw the graph of $1 + 2\sin\theta$.
- ④ a On the same axes drawn for question 3, draw the graph of $3\cos\frac{1}{2}\theta$.
- b Use your graph to estimate the value of θ for which $1 + 2\sin\theta = 3\cos\frac{1}{2}\theta$ in the range of $0^\circ \leq \theta \leq 360^\circ$.

- ⑤ Use the method in question 4 to find the values of θ between 0° and 360° for which:

a $\sin\theta = \frac{1}{2} - \cos\theta$

b $3\cos 2\theta = \sin\theta - 2$.

- ⑥ a Copy and complete Table 6.10 to give values of $y = \sin 2\theta - \cos\theta$.

θ	2θ	$\sin 2\theta$	$\cos\theta$	$y = \sin 2\theta - \cos\theta$
0°	0°	0	1	-1
30°	60°	0.87	0.87	0
60°	120°	0.87	0.50	0.37
90°	180°	0	0	0
120°				
150°				
180°				

Table 6.10

- b Using a scale of 2 cm to 30° on the horizontal axis and 5 cm to 1 unit on the vertical axis, draw the graph of $y = \sin 2\theta - \cos\theta$ for $0^\circ \leq \theta \leq 180^\circ$.
- c Use your graph to find:
- the solution of the equation $\sin 2\theta - \cos\theta = 0$, correct to the nearest degree
 - the maximum value of y , correct to one decimal place.

[WAEC]

SUMMARY

- 1 The following table shows the sign (whether positive or negative) and magnitude (size) of the sine, cosine and tangent of general angles θ in each of the four quadrants.

Quadrant	Range of values of θ	Trigonometrical ratio
1st	$0^\circ < \theta < 90^\circ$	$\sin\theta = +\sin\theta$ $\cos\theta = +\cos\theta$ $\tan\theta = +\tan\theta$
2nd	$90^\circ < \theta < 180^\circ$	$\sin\theta = +\sin(180^\circ - \theta)$ $\cos\theta = -\cos(180^\circ - \theta)$ $\tan\theta = -\tan(180^\circ - \theta)$
3rd	$180^\circ < \theta < 270^\circ$	$\sin\theta = -\sin(\theta - 180^\circ)$ $\cos\theta = -\cos(\theta - 180^\circ)$ $\tan\theta = +\tan(\theta - 180^\circ)$
4th	$270^\circ < \theta < 360^\circ$	$\sin\theta = -\sin(360^\circ - \theta)$ $\cos\theta = +\cos(360^\circ - \theta)$ $\tan\theta = -\tan(360^\circ - \theta)$

- 2 Sketches of sine, cosine and tangent curves are shown in Fig. 6.5 a and b and Fig. 6.6, respectively.

- 3 The graphs of $\sin\theta$, $\cos\theta$ and $\tan\theta$ can be drawn as follows:

- using trigonometric tables, prepare a table of values of trigonometric ratios for values of θ from 0° to 360° at equal intervals
- on appropriate scales, plot the points and draw the curve.

See Examples 4, 5 and 6.

OBJECTIVES

By the end of this chapter you should be able to:

- 1 Distinguish between great circles and small circles on a spherical surface
- 2 Recall and use the definitions of lines of longitude (including the Greenwich meridian) and latitude (including the equator) on the surface of the Earth
- 3 Determine and sketch the position of a point on the Earth's surface in terms of its latitude and longitude, e.g. (14°N , 26°E) or (37°S , 105°W)
- 4 Calculate the distance between two points on a great circle (meridian or equator)
- 5 Calculate the distance between two points on a parallel of latitude
- 6 Calculate the speed of a point on the surface of the Earth due to the Earth's rotation
- 7 Compare great circle and small circle routes on the surface of the Earth.

Teaching and learning materials

Teacher: Globe of the Earth; a skeletal model of a sphere with at least three great circles (including the equator) and at least one circle of latitude

7.1 Great circles and small circles

Fig. 7.1 **a** shows a spherical orange. The top has been cut off.

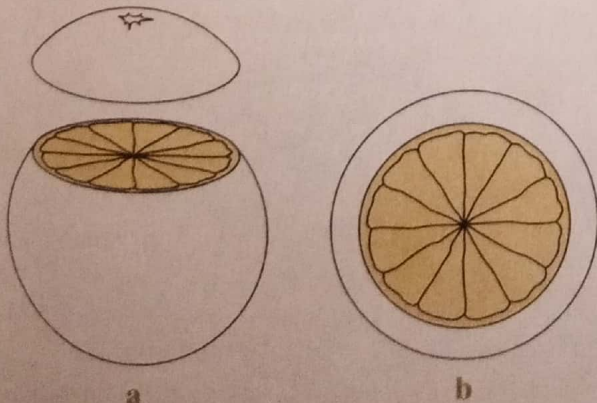


Fig. 7.1

The shape of the cross-section is a circle (Fig. 7.1 **b**). In Fig. 7.2 **a**, another cut has been made closer to the centre of the orange; the cross-section is a larger circle.

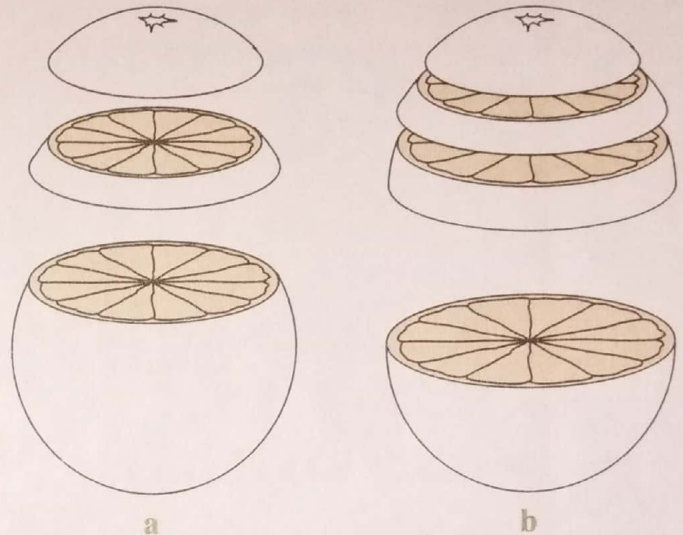


Fig. 7.2

In Fig. 7.2 **b**, a third cut has been made through the *centre* of the orange. The cross-section gives the largest possible circle that can be obtained from the orange. This is known as a **great circle**. The radius of a great circle of a sphere is equal to the radius of the sphere. Since there are any number of ways of making a straight cut through the centre of a sphere, there are any number of great circles of a sphere.

The circles of the cross-sections in Fig. 7.1 and Fig. 7.2 **a** are known as **small circles**. The radius of a small circle is less than the radius of the sphere.

Example 1

Fig. 7.3 shows a ball, centre O , resting on a circular hole in a piece of cardboard (shaded). If the radius of the ball is 3 cm and its centre is 2 cm above the plane of the cardboard, calculate the area of the circular hole.

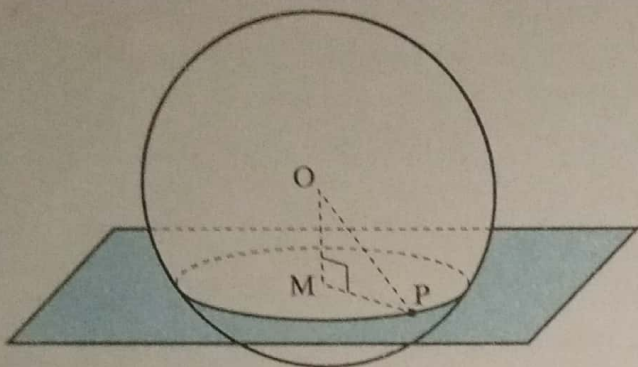


Fig. 7.3

Fig. 7.4 is a vertical section of the ball showing O, the centre of the ball, and M, the centre of the hole (a small circle).

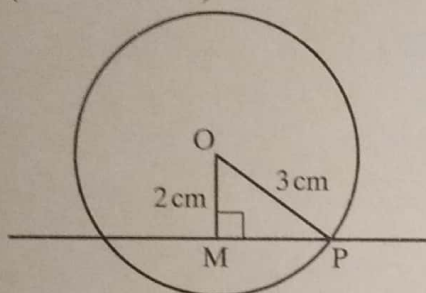


Fig. 7.4

In Fig. 7.3 and Fig. 7.4, $\overline{MP}$ is a radius of the circular hole.

$$|MP|^2 = 3^2 - 2^2 \text{ (Pythagoras)}$$

$$= 9 - 4 = 5$$

$$\Rightarrow |MP| = \sqrt{5} \text{ cm}$$

$$\begin{aligned} \text{Area of the hole} &= \pi \times |MP|^2 \text{ cm}^2 \\ &= \pi \times (\sqrt{5})^2 \text{ cm}^2 \\ &= 5\pi \text{ cm}^2 \end{aligned}$$

Exercise 7a

- ① Fig. 7.5 a shows a spherical peeled orange. Are the circles partly shown in Fig. 7.5 a great circles or small circles?

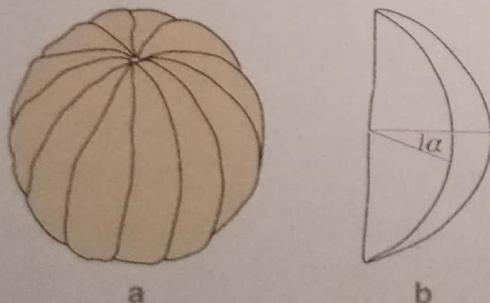


Fig. 7.5

- ② The orange in Fig. 7.5 a has 12 sectors, each assumed to be identical. What is the angle between the semicircular faces of each sector (i.e. α in Fig. 7.5 b)?

- ③ Fig. 7.6 shows a small circle and part of a great circle drawn on the surface of a sphere. Which length is smaller, arc AXB or arc AYB?

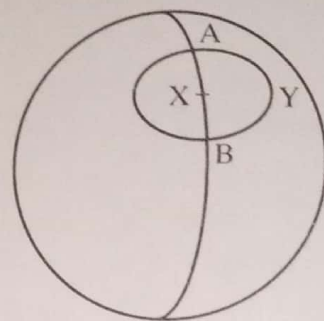


Fig. 7.6

- ④ If the ball in Fig. 7.3 is of radius 10 cm and if O is 6 cm above the cardboard, calculate:
- the radius of the circular hole
 - the area of the hole in terms of π .
- ⑤ Fig. 7.7 shows a vertical section of the orange in Fig. 7.1

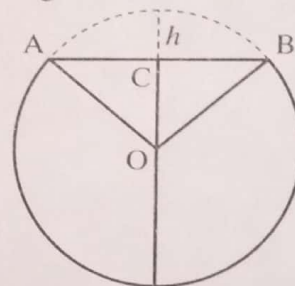


Fig. 7.7

The diameter of the small circle is 6 cm and the diameter of the orange is 10 cm.

- What is $|OA|$?
- What is $|CA|$?
- Calculate $|OC|$.
- Hence find h , the height of the slice that has been cut off.

7.2 The Earth, latitude and longitude

The Earth is approximately spherical in shape, its radius varying between 6 360 km and 6 380 km. Fig. 7.8 is a view of a model of the Earth.

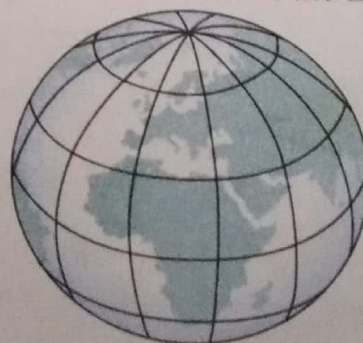


Fig. 7.8

Notice the circles drawn on the surface of the Earth. These are imaginary lines used for giving the position of points on the Earth's surface. They are called lines of **latitude** and **longitude**.

Longitude

The Earth rotates about its **polar axis**. This is a straight line through the centre of the Earth joining the North and South Poles. In Fig. 7.9, NOS is the polar axis.

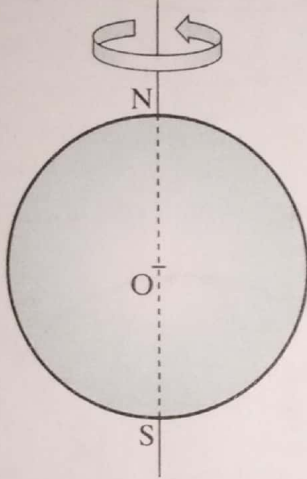


Fig. 7.9

If the Earth is cut by planes through $\overline{NS}$, the circles formed on the surface are great circles called **meridians**. Fig. 7.10 a shows some meridians (NBS, NFS, NGS, NHS and NQS).

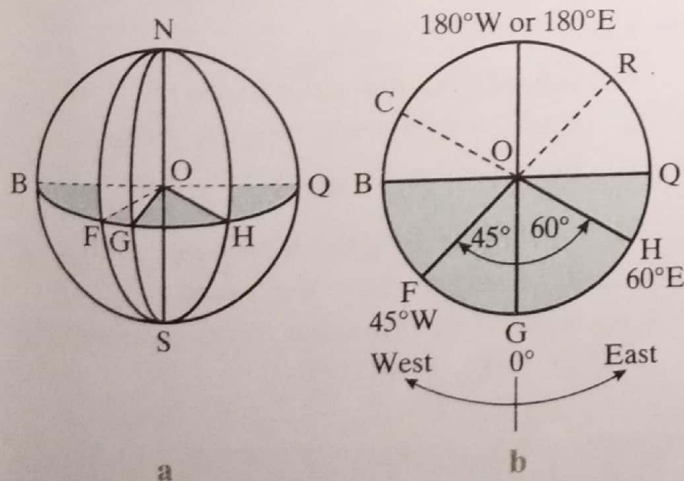


Fig. 7.10

In Fig. 7.10 a, NGS represents the meridian which passes through Greenwich in London. The **Greenwich Meridian** is used as the standard from which the positions of other meridians are measured in degrees east or west.

Fig. 7.10 b is the plan view of the meridians in Fig. 7.10 a. The **longitude** of H is the angle that the half-circle NHS makes with the half-circle

NGS, i.e. $\widehat{HOG}$, which is 60° in Fig. 7.10 b. Hence NHS is the **line of longitude** 60°E . Similarly, NFS is the line of longitude 45°W . Longitudes vary from 180°E to 180°W .

Latitude

If the Earth is cut by planes perpendicular to the polar axis, the circles formed on its surface are called **lines of latitude**. The line of latitude whose centre is also the centre of the Earth is a great circle called the **equator**. In Fig. 7.11, O is the centre of the Earth and the equator has been drawn.

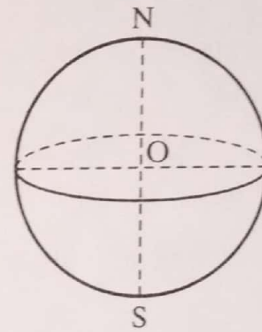


Fig. 7.11

All of the other lines of latitude are small circles, often called **parallels of latitude** since they are parallel to each other. Fig. 7.12 a shows the equator and two parallels of latitude.

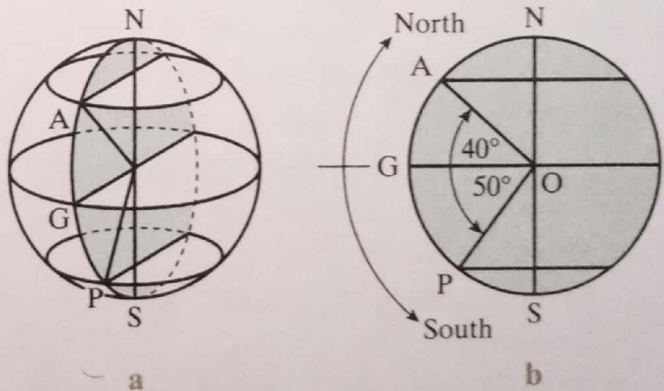


Fig. 7.12

The latitude of a point is measured in degrees north or south of the equator. Fig. 7.12 b is a cross-section in the plane NAGPS. The latitude of A is the angle that $\overline{OA}$ makes with $\overline{OG}$. Hence A is latitude 40°N . Similarly P is latitude 50°S . Latitudes vary from 90°N to 90°S .

The position of a point on the Earth's surface can be completely defined by its latitude and longitude.

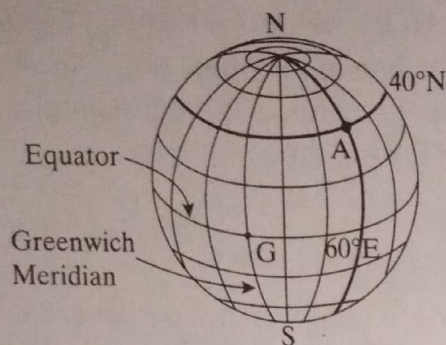


Fig. 7.13

Fig. 7.13 shows the point A(40°N, 60°E). There is only one such place on the surface of the Earth.

Exercise 7b

- ① In Fig. 7.10, state the longitude of
a B **b** Q **c** R **d** C.
- ② In Fig. 7.10, what is the difference in longitude between H and F?
- ③ How many parallels of latitude have a radius of 2 500 km?
- ④ In Fig. 7.12 **b**, what is the difference in latitude between
a A and P **b** A and S?
- ⑤ Use an atlas to find the country in which A(40°N, 60°E), Fig. 7.13, is situated.
- ⑥ Use an atlas to find, to the nearest degree, the latitude and longitude of your town or village.
- ⑦ X and Y are points on the Earth's surface at opposite ends of a diameter through the centre of the Earth.
a If the longitude of X is 19°E, what is the longitude of Y?
b If the latitude of X is 52°N, what is the latitude of Y?
- ⑧ In Fig. 7.14, G is the point where the Greenwich Meridian crosses the equator. Lines of latitude 70°N and 30°S, and lines of longitude 80°E and 40°W are given. State the latitude and longitude of the following points.
a P **b** Q **c** R
d X **e** Y **f** Z

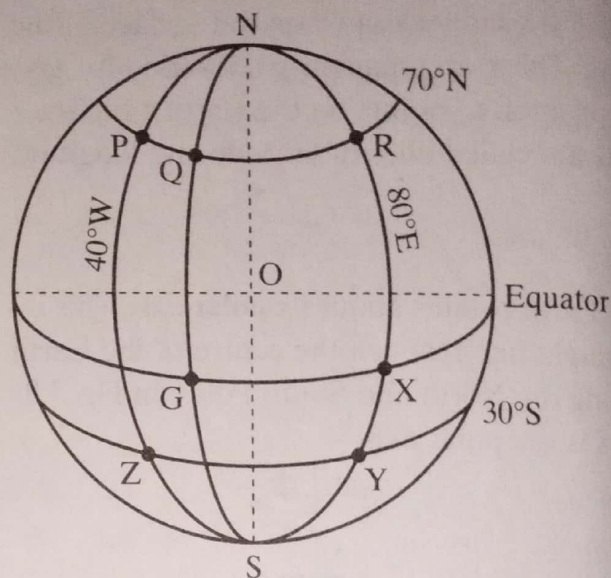


Fig. 7.14

- ⑨ Make a rough sketch of Fig. 7.15.

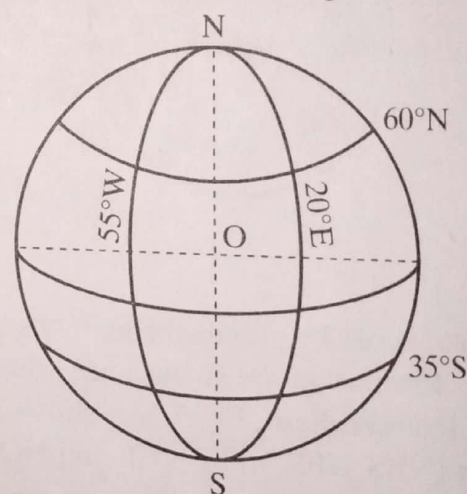


Fig. 7.15

On your sketch, mark the following points.

- a** M(35°S, 55°W) **b** P(60°N, 55°W)
c Q(60°N, 20°E) **d** R(35°S, 20°E)
- ⑩ In the 18th century, the length of $\frac{1}{4}$ of the Earth's circumference was taken to be 10 000 km. Use this value to calculate the radius of the Earth to 3 s.f.

7.3 Distances along great circles

In this section, only great circle distances along the equator and along lines of longitude are considered. Note, however, that it is possible for there to be great circles on the Earth's surface other than these. (See Fig. 7.25.)

The radius of the Earth, R , is often given as 6 400 km, which is correct to 2 s.f. The circumference of the Earth, $2\pi R$, is 40 000 km.

which is correct to 3 s.f. (See question 10 on the previous page.) Either of these values may be used in the remainder of the chapter, unless you are told otherwise.

Example 2

The positions of Libreville (Gabon) and Kampala (Uganda) to the nearest degree are $(0^\circ\text{N}, 9^\circ\text{E})$ and $(0^\circ\text{N}, 32^\circ\text{E})$, respectively. Calculate their distance apart along the equator.

Note that latitude 0°N is the equator.

Fig. 7.16 *a* shows Libreville, L, and Kampala, K, on the equator. NGS is the Greenwich Meridian.

Fig. 7.16 *b* shows the cross-section through the equator.

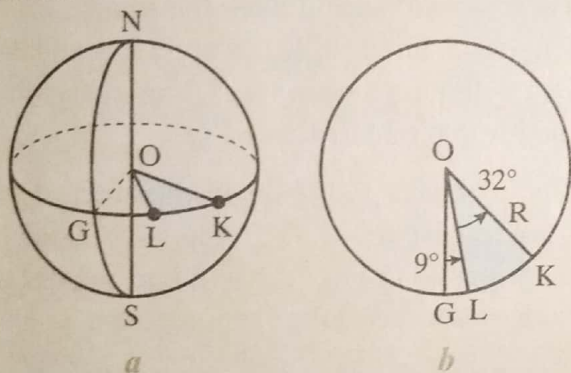


Fig. 7.16

From Fig. 7.16 *b*,
 $\widehat{LOK} = 32^\circ - 9^\circ = 23^\circ$

$\Rightarrow$ arc LK

$$= \frac{23}{360} \times 2\pi R$$

$$= \frac{23 \times 2 \times \pi \times 6400}{360} \text{ km}$$

$$= \frac{23 \times \pi \times 6400}{180} \text{ km}$$

$$= 2568 \text{ km}$$

$$\approx 2600 \text{ km to 2 s.f.}$$

Since the given data are correct to 2 s.f., it is appropriate to round off the answer to 2 s.f. Notice that the answer assumes that the shortest distance between Libreville and Kampala is required. The distance along the equator could also be given as 37 400 km $(40\,000 - 2\,600)$; this is the distance going the long way round.

Example 3

Sidi Barrani (Egypt) and Gaborone (Botswana) both lie on longitude 25.9°E . Their latitudes are 31.6°N and 24.8°S , respectively. Calculate the shortest distance between the towns.

Fig. 7.17 *a* shows the positions of Sidi Barrani, B, and Gaborone, G. Fig. 7.17 *b* shows the cross-section NBGS. OE represents the radius of the Earth at the equator.

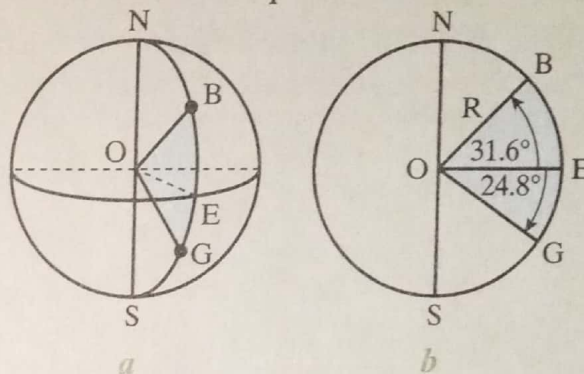


Fig. 7.17

From Fig. 7.17 *b*,

$$\widehat{BOG} = 31.6^\circ + 24.8^\circ = 56.4^\circ$$

$$\Rightarrow \text{arc BG} = \frac{56.4}{360} \times 2\pi R$$

$$= \frac{56.4}{360} \times 40\,000 \text{ km}$$

$$= \frac{56.4 \times 1000}{9} \text{ km}$$

$$\approx 6\,270 \text{ km}$$

In this example, the data are given to 3 s.f. So it is appropriate to round the answer to 3 s.f.

Example 4

P and Q are two places on the equator. Their longitudes are 173°E and 75°W , respectively.

a What is the shortest distance between them along the equator?

b How far is Q from the South Pole?

(Take π to be $\frac{22}{7}$ and the radius of the Earth to be 6 400 km.)

Fig. 7.18 *a* and Fig. 7.18 *b* show a view and the plan of P and Q, respectively.

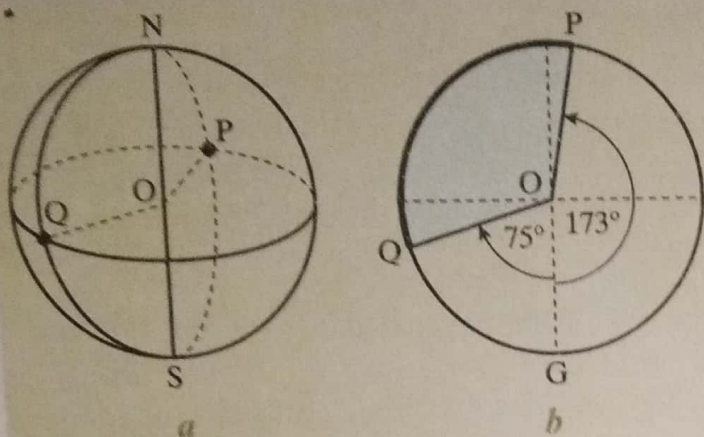


Fig. 7.18

a In Fig. 7.18 *b*, the shortest distance between P and Q is the shorter arc PQ where non-reflex $\widehat{POQ} = 360^\circ - (173^\circ + 75^\circ) = 112^\circ$

$$\text{Shorter arc PQ} = \frac{112}{360} \times 2\pi R$$

$$= \frac{112 \times 2 \times 22 \times 6400}{360 \times 7} \text{ km}$$

$$= \frac{16 \times 22 \times 6400}{180} \text{ km}$$

$$= 12516 \text{ km (calculator)}$$

$$= 13000 \text{ km (2 s.f.)}$$

b In Fig. 7.18 *a*, arc QS gives the distance of Q from the South Pole.

$$\text{Arc QS} = \frac{90}{360} \times 2\pi R = \frac{1}{4} \times 2\pi R$$

$$= \frac{11 \times 6400}{7} \text{ km}$$

$$= 10057 \text{ km (calculator)}$$

$$= 10000 \text{ km (to 2 s.f.)}$$

Note: In Example 4, the distance of any point on the equator from either pole is one-quarter of the circumference of the Earth. Also, in this example, it would have been more accurate (and easier!) to use the value 40 000 km for $2\pi R$. However, since the question required the use of $R = 6400$ km, this value was used. (Please note this point in your examination's rubric. See page 240.)

Figs 7.16, 7.17 and 7.18 were of considerable help when solving Examples 2, 3 and 4. Each contained a perspective view and a cross-section which showed the required arc clearly. Always draw similar sketches when solving latitude and longitude problems.

Exercise 7c

Use the values 6 400 km for R or 40 000 km for $2\pi R$ as appropriate. Take $\log \pi$ to be 0.4971.

- 1 In Fig. 7.19, points X and Y lie on the same meridian. Their latitudes are 15°N and 36°N . Find the distance XY measured along the meridian.

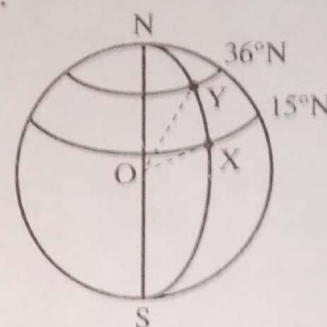


Fig. 7.19

- 2 Two places P and Q have the same longitude. Their latitudes are 15°N and 36°S . Draw diagrams as in Fig. 7.17 and calculate the great-circle distance PQ.
- 3 Two places G and H lie on the equator, and have longitudes of 92°E and 57°E . Draw diagrams as in Fig. 7.16 and find the distance GH, measured along the equator.
- 4 Two places P and Q with longitudes of 152°E and 171°W , both lie on the equator. Draw diagrams as in Fig. 7.18 and find the shortest distance between P and Q on the Earth's surface.
- 5 The positions of Abuja (Nigeria) and Bonn (Germany) to the nearest degree are $(9^\circ\text{N}, 7^\circ\text{E})$ and $(51^\circ\text{N}, 7^\circ\text{E})$, respectively. Use $R = 6400$ km to calculate their distance apart to 2 s.f.
- 6 The Sao Tome and Galapagos Islands lie on the equator on longitudes 7°E and 90°W , respectively. If the circumference of the Earth is 40 000 km, find their distance apart to 3 s.f.
- 7 By road, Opobo Town and Katsina are 1312 km apart. Their positions are $(4.6^\circ\text{N}, 7.5^\circ\text{E})$ and $(13.0^\circ\text{N}, 7.5^\circ\text{E})$, respectively.
 - a Calculate their great-circle distance apart.
 - b Explain why there is a difference between your result and the road distance.

- 8 Find the distance, measured along a meridian, of the South Pole from any point on the parallel of latitude 65°S .
- 9 Two places X and Y on the equator are on longitudes 67°E and 123°E , respectively.
- What is the distance between them along the equator?
 - How far is X from the North Pole?
(Take $\pi = \frac{22}{7}$ and the radius of the Earth as 6 400 km). [WAEC]
- 10 Points Z($n^\circ\text{N}$, 0°) and W(27°S , 0°) lie on the Greenwich Meridian. If Z is 8 000 km from the North Pole, calculate:
- the value of n to the nearest integer
 - the distance ZW along the Greenwich Meridian.
- (Use the values 3.14 for π and 6 400 km for the radius of the Earth. In part b use your integer value of n .)
- 11 Show that the points (20°N , 20°E) and (60°N , 160°W) lie on the same great circle. Find their great-circle distance apart.
- 12 Two places are 264 km apart on the same line of longitude. Calculate the difference between their latitudes.

7.4 Distances along parallels of latitude

Radius of a parallel of latitude

In Fig. 7.20, the latitude of point P is θ . C is the centre of the circle of latitude on which P lies.

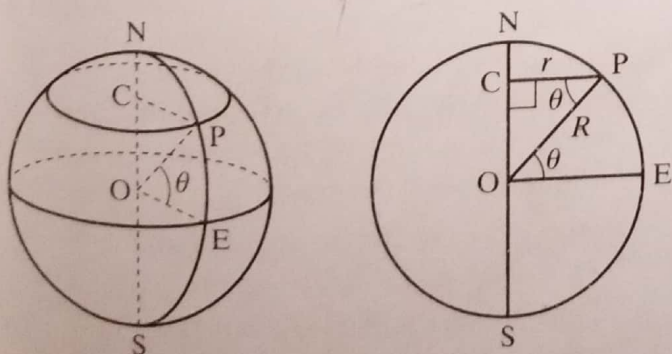


Fig. 7.20

Let r be the radius of the parallel of latitude through P, then, in $\triangle PCO$:

$$\widehat{CPO} = \theta \text{ (alt. angles } \overline{CP} \parallel \overline{OE})$$

$$|OP| = R \text{ (radius of Earth)}$$

$$\Rightarrow \cos \theta = \frac{r}{R}$$

$$\Rightarrow r = R \cos \theta$$

The relationship $r = R \cos \theta$ is true for all parallels of latitude that are θ north or south.

Example 5

Find the distance, measured along the parallel of latitude, between two points whose latitudes are both 56°N , and whose longitudes are, respectively, 23°E and 17°W .

In Fig. 7.21, A is the point (56°N , 23°E) and B is the point (56°N , 17°W). C is on the polar axis and is the centre of the parallel of latitude 56°N . Fig. 7.21 b – is the plan view of the parallel of latitude.

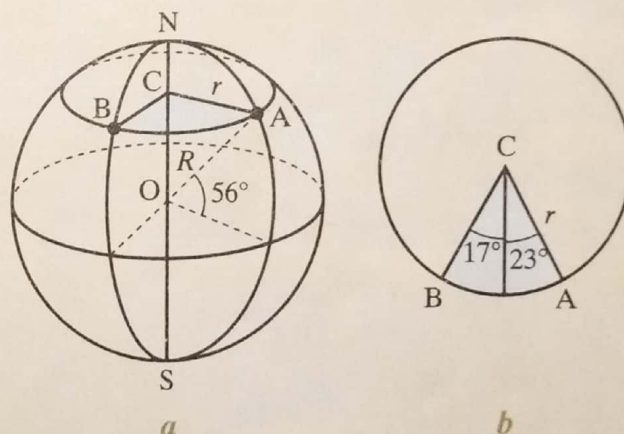


Fig. 7.21

In $\triangle ACO$, $r = R \cos 56^\circ$.

In Fig. 7.21 b,

$$\widehat{ACB} = 23^\circ + 17^\circ = 40^\circ$$

$$\begin{aligned} \Rightarrow \text{arc AB} &= \frac{40}{360} \times 2\pi r \\ &= \frac{40}{360} \times 2\pi R \cos 56^\circ \end{aligned}$$

$$\text{But } 2\pi R = 40\,000 \text{ km}$$

$$\begin{aligned} \Rightarrow \text{arc AB} &= \frac{40}{360} \times 40\,000 \times \cos 56^\circ \text{ km} \\ &= \frac{40\,000 \times 0.5592}{9} \text{ km} \\ &\approx 2\,490 \text{ km} \end{aligned}$$

Example 6

Find the parallel of latitude in the northern hemisphere along which a journey of 640 km makes a change of 8° in longitude.

Let the required parallel of latitude be $\theta^\circ\text{N}$.
Fig. 7.22 shows the data in the question.

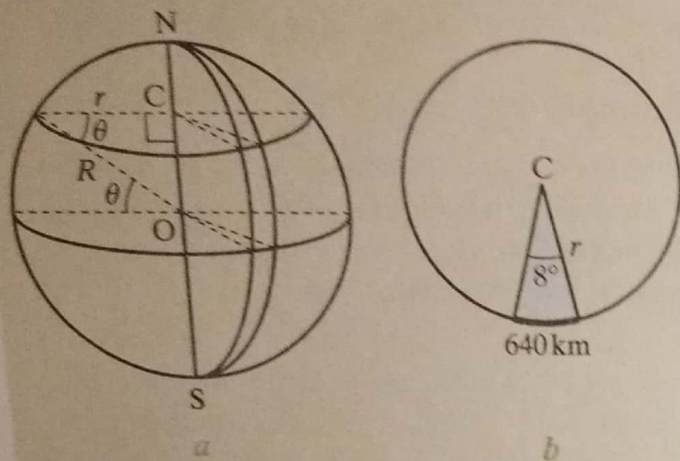


Fig. 7.22

In Fig. 7.22 a, $r = R\cos\theta$.

In Fig. 7.22 b:

$$\begin{aligned} 640 &= \frac{8}{360} \times 2\pi r \\ &= \frac{8}{360} \times 2\pi R\cos\theta \\ &= \frac{8}{360} \times 40\,000 \cos\theta \\ \cos\theta &= \frac{640 \times 360}{8 \times 40\,000} = 0.72 \\ \theta &= 43.95 \\ &= 44^\circ \text{ to the nearest degree.} \end{aligned}$$

The parallel of latitude is 44°N .

Exercise 7d

Unless told otherwise, use the values 6 400 km for R or 40 000 km for $2\pi R$. Take $\log\pi$ to be 0.4971.

- ① A geographical globe has a radius of 18 cm. Find the radius of the circle formed by the parallel of latitude 56°S .
- ② On a globe of radius 60 cm, find the length in cm of the parallel of latitude 67°N .
- ③ Find the distance of a place with latitude 68°S from the polar axis of the Earth.

- ④ Find the length of the parallel of latitude 42°N .
- ⑤ Find the length of the Tropic of Capricorn, latitude 23.5°S .
- ⑥ Find the distance, measured along the parallel of latitude between two places with the same latitude, 18°S , and with longitudes 96°E and 57°E .
- ⑦ Two places are on the parallel of latitude 77°N . Their longitudes are 154°E and 142°W . Find their distance apart measured along the parallel of latitude.
- ⑧ Find the distance apart, measured along the parallel of latitude, of two places which both have latitude 63°N , and whose longitudes differ by 1° .
- ⑨ Find the distance, measured along the parallel of latitude, between Monrovia (6.3°N , 10.8°W) and Benin City (6.3°N , 5.7°E).
- ⑩ Find the distance between Bida (9°N , 6.0°E) and Addis Ababa (9°N , 38.8°E) measured along the parallel of latitude.
- ⑪ Havana and Canton are both on latitude 23°N , and their longitudes are 82.25°W and 113.25°E , respectively. Find their distance apart measured along the parallel of latitude.
- ⑫ X is a point on latitude 28.6°N and longitude 105.7°W . Y is on the same latitude as X but lies on longitude 34.3°E . Calculate the distance between X and Y along latitude 28.6°N .
(Assume that the Earth is a sphere of radius 6 370 km.) [WAEC]
- ⑬ Two points M and N on the surface of the Earth are given by their latitudes and longitudes as M(50°S , 15°E) and N(50°S , 75°E). Calculate
 - a the radius of the parallel of latitude on which M and N lie
 - b the distance MN measured along the parallel of latitude.(Take the radius of the Earth to be 6 400 km.) [WAEC]

- 14 Find the parallel of latitude along which a journey of 166 km makes a change of 4° in longitude. (Give your answer to the nearest degree.)
- 15 The position of a point A is $(41^\circ\text{N}, 18^\circ\text{E})$. Point B is 8000 km due west of A. Find the longitude of B to the nearest whole degree.

Example 7

A point X is on latitude 32°N . Calculate the speed of X due to the rotation of the Earth. The Earth completes one revolution in 24 hours. Hence X travels a distance equal to the circumference of its circle of latitude in 24 hours (Fig. 7.23).

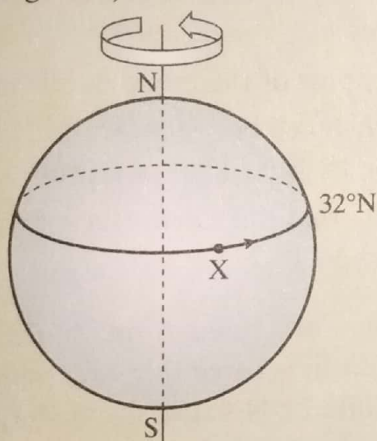


Fig. 7.23

$$\begin{aligned}\text{Circumference of parallel of latitude } 32^\circ\text{N} &= 2\pi R \cos 32^\circ \\ &= 40\,000 \times 0.8480 \text{ km} \\ \text{Speed of X} &= \frac{40\,000 \times 0.8480}{24} \text{ km/h} \\ &= 1\,410 \text{ km/h}\end{aligned}$$

Example 8

An aeroplane leaves a point P($21^\circ\text{N}, 32^\circ\text{E}$) and flies due south at 850 km/h. Find its position Q after 4 hours. (Assume that the Earth is a sphere of radius 6 400 km.)

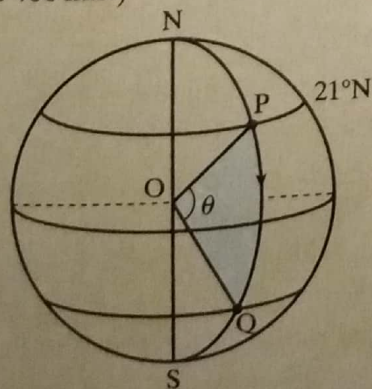


Fig. 7.24

Distance travelled in 4 hours

$$= 850 \times 4 \text{ km}$$

$$= 3\,400 \text{ km}$$

With the lettering used in Fig. 7.24:

$$3\,400 = \frac{\theta}{360} \times 2\pi R$$

$$= \frac{\theta \times 2 \times \pi \times 6\,400}{360}$$

$$\theta = \frac{3\,400 \times 360}{2 \times \pi \times 6\,400}$$

$$= \frac{1\,530}{16\pi}$$

$$= 30 \text{ to } 2 \text{ s.f.}$$

$$\text{Latitude of Q} = 21^\circ\text{N} - 30^\circ$$

$$= 9^\circ\text{S}$$

Q is $(9^\circ\text{S}, 32^\circ\text{E})$.

No	Log
1 530	3.184 7
16	1.204 1
π	0.497 1
	1.701 2
30.45	1.483 5

7.5 Great-circle routes

In the geometry of the Earth, the equator and the meridians of longitude are the most common great circles. However, there is an infinite number of great circles on the surface of a sphere (since there are infinitely many ways of cutting a sphere in half by a plane through its centre).

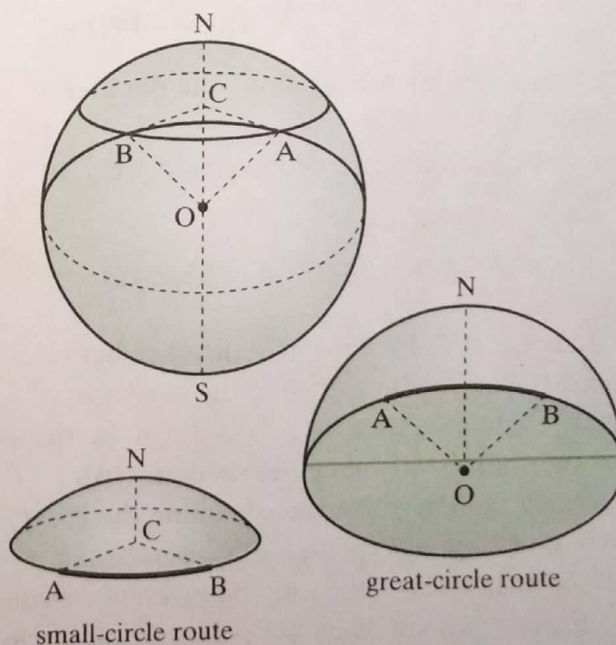


Fig. 7.25

Fig. 7.25 is a sketch showing points A and B on both a circle of latitude, centre C, and on a great circle, centre O. Indeed, any two points on the surface of the Earth lie on a great circle.

Example 9

A and B are two points on latitude 70°N . Their longitudes are 62°W and 118°E , respectively.

Find the distance from A to B along:

- a a great circle b a parallel of latitude.

- a The longitudes of A and B differ by 180° ($62^\circ + 118^\circ = 180^\circ$). Hence A and B lie on the same meridian, a great circle passing through the North Pole. This is shown in Fig. 7.26.

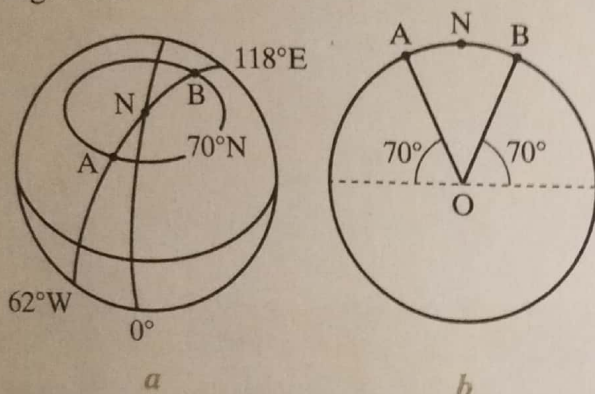


Fig. 7.26

In Fig. 7.26 b, O is the centre of the Earth.

$$\angle AOB = 180^\circ - 2 \times 70^\circ = 40^\circ$$

$$\begin{aligned} \Rightarrow \text{arc ANB} &= \frac{40}{360} \times 2\pi R \\ &= \frac{40 \times 40\,000}{360} \text{ km} \approx 4\,440 \text{ km} \end{aligned}$$

- b Distance from A to B along the parallel of latitude
- $$\begin{aligned} &= \frac{1}{2} \times \text{circumference of the small circle, } 70^\circ\text{N} \\ &= \frac{1}{2} \times 2\pi R \cos 70^\circ \\ &= \frac{1}{2} \times 40\,000 \times 0.3420 \text{ km} = 6\,840 \text{ km} \end{aligned}$$

In Example 9, the great-circle distance between A and B is much shorter than the small-circle distance between the two points. It can be shown that **the shortest distance between any two points on the Earth's surface is along the great circle that connects them**. Airline companies use this fact when navigating between places on the Earth's surface. They always use great-circle routes where possible.

Exercise 7e

Unless told otherwise, use the value 40 000 for $2\pi R$. Take $\log \pi$ to be 0.4971.

- ① A point P lies on the Tropic of Cancer, 23.5°N . Calculate the speed of P in km/h due to the rotation of the Earth.
- ② P is due north of Q and the distance PQ along the surface of the Earth is 1 120 km. If the latitude of P is 3.2°N , calculate the latitude of Q.
- ③ Two places A and B are 480 km apart and A is due south of B. If the latitude of A is 47.8°N , find the latitude of B.
- ④ Point A (latitude 28°N , longitude 33°E) and point B (latitude 28°N , longitude 27°W) are two points on the surface of the Earth. Calculate:
 - a the radius of the circle of latitude 28°N
 - b the distance between A and B measured along their parallel of latitude.
 (Assume that the Earth is a sphere of radius 6 400 km.) [WAEC]
- ⑤ The speed of a point A due to the rotation of the Earth is twice that of a point B. If A is on latitude 22°N , calculate the latitude of B.
- ⑥ Ilorin is at 8.5°N , 4.6°E and Freetown is at 8.5°N , 13.3°W . Calculate:
 - a their distance apart, measured along the parallel of latitude
 - b their distance from the equator
 - c their speed, in km/h, due to the rotation of the Earth.
- ⑦ The latitude and longitude of a point P are 50°N , 32°W and of another point Q are 50°N , 48°E . Calculate, to the nearest 100 km:
 - a the radius of the circle of latitude through P and Q
 - b the distance PQ along the parallel of latitude
 - c the speed due to the rotation of the Earth in km/h of the point P, assuming that the Earth makes a complete rotation in 24 hours.
 (Take the radius of the Earth to be 6 400 km.) [WAEC]